Supplement to *What Pre-Trends Cannot See: Testing for Treatment-Induced Recording Change with Few Clusters*

Ethan Xie

This supplement contains the proofs of every result in the paper, the simulation and application evidence behind the numbers the paper reports, the extended discussion of the related and applied literature, and the figure of exact size against the number of restrictions. Numbering follows the paper: a reference to Theorem 1 here is Theorem 1 there.

S1 Proofs

Proof of Lemma 1. Write s_1 for the structure of the statement: latent outcomes Y^* satisfying parallel trends, recording paths m_c with $m_c(\ell) = 0$ for $\ell < 0$, and unit by category terms μ_{ic}, period by category terms ϕ_{tc}, recording noise u_{itc} and cohort labels G_i. By Assumption 1 the observed panel under s_1 is

$$Y_{itc} = Y^*_{itc} + \mu_{ic} + \phi_{tc} + m_c(\ell_{it})D_{it} + u_{itc}. \tag{S1.1}$$

Construct s_2 on the same probability space by leaving μ, ϕ, u and G unchanged, setting the recording paths to zero, and replacing the latent untreated potential outcome by

$$Y^\dagger_{itc}(\infty) = Y^*_{itc}(\infty) + m_c(t-g)\mathbf{1}\{t \geq g\} \qquad \text{on } \{G_i = g\}, \tag{S1.2}$$

with treated potential outcomes shifted by the same increment, so that latent treatment effects are unchanged. Since s_2 has no recording response, its observed panel is $Y_{itc} = Y^\dagger_{itc} + \mu_{ic} + \phi_{tc} + u_{itc}$, and substituting (S1.2) and $\ell_{it} = t - G_i$ returns (S1.1) cell by cell, pointwise on the underlying sample space. The two structures therefore induce the same joint law of $\{(Y_{itc})_{t,c}, G_i\}_{i \leq N}$, not merely the same moments.

For the deviation induced by s_2, note that s_1 satisfies latent parallel trends, so any departure under s_2 is the increment in (S1.2). Differencing it across t and across cohorts in the definition of $\Delta_c(g,t)$ gives $\Delta_c(g,t) = m_c(t-g)\mathbf{1}\{t \geq g\}$, which is (3.2). It vanishes for $t < g$ because $m_c(\ell) = 0$ for $\ell < 0$, which is the no anticipation restriction in the recording layer. □

Proof of Corollary 1. Let φ be measurable with respect to $\sigma(\{(Y_{itc})_{t,c}, G_i\}_{i \leq N})$. By Lemma 1 the law of that array is the same under s_1 and s_2, so $E_{s_1}[\varphi] = E_{s_2}[\varphi]$. Under s_2 the deviation $\Delta_c(g,t)$ is identically zero for $t < g$, so s_2 lies in the null of any test whose null hypothesis restricts pre treatment deviations. If φ has size α over that null then $E_{s_2}[\varphi] \leq \alpha$, with equality when the test is similar. Hence $E_{s_1}[\varphi] \leq \alpha$, and equal to α for a similar test, although s_1 carries a recording response of arbitrary magnitude. The same argument applies to placebo in time statistics and to leads and lags coefficient vectors, which are also functions of the observed panel and the cohort labels. □

Proof of Proposition 1. Adopt the normalisation $\delta_{-1} = 0$ and let the pre period first differences be $\delta_{s+1} - \delta_s$ for $s \leq -2$, which are the differences a relative magnitudes calibration reads. By Lemma 1 the deviation equivalent to the recording response is $\delta_\ell = m_c(\ell)\mathbf{1}\{\ell \geq 0\}$, so $\delta_\ell = 0$ for every $\ell \leq -1$ and every pre period first difference is exactly zero. Hence $\max_{s \leq -2} |\delta_{s+1} - \delta_s| = 0$,

and the defining constraint of (3.3) reads $|\delta_{\ell+1} - \delta_\ell| \le \bar{M} \cdot 0 = 0$ for every $\ell \ge -1$ and every $\bar{M} \ge 0$. Together with $\delta_{-1} = 0$ this forces $\delta_\ell = 0$ for all ℓ, so $\Delta^{\mathrm{RM}}(\bar{M}) = \{0\}$ irrespective of $\bar{M}$.

The identified set for the target under a deviation class Δ is $\{\, l'\beta_{\mathrm{post}} - l'\delta_{\mathrm{post}} : \delta \in \Delta \,\}$ with δ_{pre} pinned at its population value. With $\Delta^{\mathrm{RM}}(\bar{M}) = \{0\}$ this is the singleton $\{\, l'\beta_{\mathrm{post}} \,\}$. By Lemma 1, $\beta_c(\ell) = \vartheta_c(\ell) + m_c(\ell)$, so the singleton is $\{\vartheta_c(\ell) + m_c(\ell)\}$ and excludes $\vartheta_c(\ell)$ whenever $m_c(\ell) \neq 0$. Since the set is a point, no value of $\bar{M}$ returns an interval containing the latent effect, and no breakdown value exists. □

Proof of Lemma 2. Under Assumption 2 the regressor block in (4.1) is common across categories after the two way within transformation, so stacking the C categories gives design matrix $[\, \mathbf{1}_C \otimes \tilde{X},\ \delta \otimes \tilde{X} \,]$ with $\delta = (\delta_1, \ldots, \delta_C)'$. Because $\mathbf{1}_C'\delta = 0$ by the centering of δ, the two blocks are orthogonal, and the normal equations separate into $(\mathbf{1}_C'\mathbf{1}_C)S\hat{\beta} = \sum_c \tilde{X}'\tilde{y}_c$ and $(\delta'\delta)S\hat{\gamma} = \sum_c \delta_c \tilde{X}'\tilde{y}_c$. Dividing by C and $\sum_c \delta_c^2$ respectively and using $b_c = S^{-1}\tilde{X}'\tilde{y}_c$ gives the stated expressions. □

Proof of Proposition 2. By (5.1) the CR1 covariance of $\hat{\gamma}$ on the post block is $\hat{V}_{PP} = \omega \sum_g \eta_{gP}^f \eta_{gP}^{f\prime}$ with $\omega = \omega_2^{-1} G/(G-1)$. The within category residual is $e_{gc} = u_{gc} - \tilde{X}_g S^{-1} \tilde{X}' u_c$ by Lemma 2, so $a_g^f = a_g - S_g S^{-1} \sum_h a_h$ and $\eta_g^f = \eta_g - S^{-1} S_g \sum_h \eta_h$. Under Assumption 3(ii), $S^{-1}S_g = I_q/G$, so $\eta_g^f = \eta_g - \bar{\eta}$, the restriction to P commutes with the correction, and $\sum_g \eta_{gP}^f \eta_{gP}^{f\prime} = (G-1)\hat{S}_P$ while $\hat{\gamma}_P - \gamma_P = G\bar{\eta}_P$. Substituting,

$$W = \hat{\gamma}_P' \hat{V}_{PP}^{-1} \hat{\gamma}_P = \omega_2\, \bar{\eta}_P' \Big[\tfrac{1}{G-1} \textstyle\sum_g (\eta_{gP} - \bar{\eta}_P)(\eta_{gP} - \bar{\eta}_P)' \Big]^{-1} \bar{\eta}_P\, G = \omega_2 T^2,$$

where the centering is the within category fitting correction and not a consequence of imposing (4.2). This is an algebraic identity and does not use Assumption 3(i).

For the distribution, Assumption 3(i) gives

$$\sqrt{G}\,\bar{\eta}_P \sim N(0, \Sigma_{PP}), \qquad (G-1)\hat{S}_P \sim W_m(\Sigma_{PP}, G-1),$$

independently, by Cochran's theorem applied to the i.i.d. Gaussian sample $\eta_{1P}, \ldots, \eta_{GP}$. Hence T^2 is Hotelling's T^2 with dimension m and $G-1$ degrees of freedom (Hotelling 1931), and the standard transformation $\frac{G-m}{(G-1)m} T^2 \sim F(m, G-m)$ holds for $G - m \ge 1$. Substituting $T^2 = W/\omega_2$ gives (5.2).

For the last claim, write $\sum_g \eta_{gP}^f \eta_{gP}^{f\prime} = E'E$ with E the $G \times m$ matrix whose gth row is $\eta_{gP}^{f\prime}$. Summing $\eta_g^f = \eta_g - S^{-1}S_g \sum_h \eta_h$ over g and using $\sum_g S_g = S$ gives $\sum_g \eta_g^f = 0$, so the columns of E are orthogonal to $\mathbf{1}_G$ and $\operatorname{rank} E \le \min(m, G-1)$. If $G - 1 < m$ the meat is singular, $\hat{S}_P$ is not invertible, and the Hotelling representation is unavailable. □

Proof of Corollary 2. By (5.2) the exact size of the nominal chi square test is $\Pr\{F(m, G-m) > (G-m)c_m/(\omega_2(G-1)m)\}$ with $c_m = \chi^2_{m,1-\alpha}$. Write $F(m, G-m) \overset{d}{=} (\chi^2_m/m)/(\chi^2_{G-m}/(G-m))$ with the two chi squares independent, so the size equals $\Pr\{\chi^2_m > c_m\, \omega_2^{-1}(G-1)^{-1}\chi^2_{G-m}\}$.

For (i), write $W \overset{d}{=} R\,\chi^2_m$ with $R = \omega_2(G-1)/\chi^2_{G-m}$ independent of χ^2_m, so the size is $\mathbb{E}[h(R)]$ with $h(x) = \Pr\{\chi^2_m > c_m/x\}$ and $h(1) = \alpha$. Put $\nu = G-1$ and $d = G-m$. At fixed m, $\mathbb{E}[R-1] = (m+1)/\nu - (1-\omega_2) + O(\nu^{-2})$ and $\mathbb{E}[(R-1)^2] = 2/\nu + O(\nu^{-2})$, using $\mathbb{E}[\chi_d^{-2k}] = \prod_{j=1}^k (d-2j)^{-1}$ for $d > 2k$ and $G(1-\omega_2) = O(1)$. The same formula gives the

third central moment of $1/\chi^2_d$ as $16/\{(d-2)^3(d-4)(d-6)\}$ for $d>6$, whence the signed third moment $\mathbb{E}[(R-1)^3]=O(\nu^{-2})$.

A second order expansion does not deliver the remainder. Bounding it by $\sup|h'''|\,\mathbb{E}|R-1|^3$ gives $O(\nu^{-3/2})$, because the absolute third moment, unlike the signed one, gains nothing from cancellation. Expand instead to third order with a fourth order Lagrange remainder. Outside $|R-1|\le 1/2$ the contribution to $\mathbb{E}[h(R)]$ is exponentially small in ν, since h is bounded by one and the chi square Chernoff bound gives $\Pr(|R-1|>1/2)\le e^{-c\nu}$ for some $c>0$. Inside, $\sup_{|x-1|\le 1/2}|h''''(x)|$ is finite and $\mathbb{E}[(R-1)^4]=3\{\mathbb{E}[(R-1)^2]\}^2+O(\nu^{-3})=12/\nu^2+O(\nu^{-3})$, so the third order term and the remainder are both $O(\nu^{-2})$.

Substituting the moments with $h'(1)=c_m f_m(c_m)$ and $h''(1)=-c_m^2 f_m'(c_m)-2c_m f_m(c_m)$, and using $f_m'(x)=f_m(x)\{(m/2-1)/x-1/2\}$ so that $c_m^2 f_m'(c_m)=c_m f_m(c_m)\{(m/2-1)-c_m/2\}$, the bracket collapses to

$$h'(1)\,\mathbb{E}[R-1]+\tfrac{1}{2}h''(1)\,\mathbb{E}[(R-1)^2]=c_m f_m(c_m)\Big[\frac{m+c_m}{2\nu}-(1-\omega_2)\Big],$$

which is the stated expansion. At the design of Section 9 with $G=51$ and $C=3$, where $\omega_2=0.9981$, the two terms in that bracket are 0.354 and 1.9×10^{-3} at $m=13$ and $\alpha=0.05$, and $G(1-\omega_2)$ is at most 0.098 over the twelve designs of Table 2 (`results/inference/anatomy_table.csv`).

For (ii), take logarithms. The event is $\log\chi^2_m-\log\chi^2_{G-m}>\log c_m-\log(\omega_2(G-1))$. With $m\to\infty$, $\log\chi^2_m$ is asymptotically normal with mean $\log m-1/m+O(m^{-2})$ and variance $2/m$, and similarly for $\log\chi^2_{G-m}$ with $G-m\to\infty$. Standardizing and using $\log c_m=\log m+\sqrt{2/m}\,z_{1-\alpha}+o(m^{-1/2})$ gives size $\to\Phi(\chi-z_{1-\alpha})$ with $\chi=\lim\sqrt{m/2}\,\log\{(G-1)/(G-m)\}$. Expanding $\log\{(G-1)/(G-m)\}=(m-1)/(G-1)+O((m/G)^2)$ gives $\chi_{m,G}=m^{3/2}/(\sqrt{2}G)+o(1)$ when $m^{5/2}/G^2\to0$. If $m/G\to r\in(0,1)$ then $\log\{(G-1)/(G-m)\}\to-\log(1-r)>0$ and $\chi_{m,G}\to\infty$, so the size tends to one. □

Proof of Lemma 3. Under Assumption 2 the two way within transformation is the same operator for every category, so relabelling the arrays gives $\hat\Sigma(\mathcal{D}_\sigma)_{ck}=\hat\Sigma_{\sigma(c)\sigma(k)}$ and hence $L(\mathcal{D}_\sigma)_{ck}=L_{\sigma(c)\sigma(k)}$. Substituting $j=\sigma(k)$ maps $\{k\ne c\}$ bijectively onto $\{j\ne\sigma(c)\}$, and substituting $(j,l)=(\sigma(k),\sigma(l))$ maps $\{k\ne l,\ k,l\ne c\}$ bijectively onto $\{j\ne l,\ j,l\ne\sigma(c)\}$, so both averages in (6.1) are evaluated at the relabelled index and $\hat a_c(\mathcal{D}_\sigma)=\hat a_{\sigma(c)}(\mathcal{D})$. The geometric mean scaling divides by $\exp\{C^{-1}\sum_j\hat a_j\}$, which is σ-invariant, so $\hat\rho_c(\mathcal{D}_\sigma)=\hat\rho_{\sigma(c)}(\mathcal{D})$. The retained set and the normalized arrays are therefore relabelled with the loadings. □

Proof of Theorem 1. The argument is the classical group invariance construction; see Lehmann and Romano (2005, Ch. 15). Order invariance of $\mathcal{T}$ gives, for every $\pi\in S_C$, $\mathcal{T}(\mathcal{D},\delta_\pi)=\mathcal{T}(\mathcal{D}_{\pi^{-1}},\delta)$ where $\mathcal{D}_{\pi^{-1}}=(\mathcal{Y}_{\pi^{-1}(1)},\dots,\mathcal{Y}_{\pi^{-1}(C)})$: the statistic is computed from the multiset $\{(\delta_{\pi(c)},\mathcal{Y}_c)\}_c$, and reindexing by $j=\pi(c)$ turns this into $\{(\delta_j,\mathcal{Y}_{\pi^{-1}(j)})\}_j$.

Under Assumption 4, $\mathcal{D}_{\pi^{-1}}\overset{d}{=}\mathcal{D}$ conditional on $\mathcal{Z}$ for each fixed π. Writing $f(\cdot)=\mathcal{T}(\cdot,\delta)$, so that $T(\pi)=f(\mathcal{D}_{\pi^{-1}})$, the substitution $\pi\mapsto\pi\sigma$ gives

$$T(\pi\sigma)=f\big(\mathcal{D}_{(\pi\sigma)^{-1}}\big)=f\big(\mathcal{D}_{\sigma^{-1}\pi^{-1}}\big)=f\big((\mathcal{D}_{\sigma^{-1}})_{\pi^{-1}}\big),$$

so $(T(\pi\sigma))_\pi$ is the same fixed measurable function of $\mathcal{D}_{\sigma^{-1}}$ that $(T(\pi))_\pi$ is of $\mathcal{D}$. By Assumption 4, $\mathcal{D}_{\sigma^{-1}}\overset{d}{=}\mathcal{D}$ conditional on $\mathcal{Z}$, whence $(T(\pi\sigma))_\pi\overset{d}{=}(T(\pi))_\pi$ for every $\sigma\in S_C$. Left translation

does not have this property, since $T(\sigma\pi) = f(\mathcal{D}_{\pi^{-1}\sigma^{-1}})$ is not a fixed function of any one relabelling of $\mathcal{D}$, and it is right translation that the counting below uses.

We bound $\Pr(\hat{p} \le \alpha)$ for the $\hat{p}$ of Definition 2 directly, which covers ties. For $\sigma \in S_C$ put $N(\sigma) = \#\{\pi' \in S_C : T(\pi') \ge T(\sigma)\}$ and $\mathcal{A} = \{\sigma : N(\sigma) \le \alpha M\}$, so the test rejects exactly on the event $\mathrm{id} \in \mathcal{A}$. If $\mathcal{A}$ is not empty, let σ^* attain the smallest value of T on it. Every $\sigma \in \mathcal{A}$ has $T(\sigma) \ge T(\sigma^*)$, so $\mathcal{A} \subseteq \{\pi' : T(\pi') \ge T(\sigma^*)\}$ and $|\mathcal{A}| \le N(\sigma^*) \le \alpha M$, hence $|\mathcal{A}| \le \lfloor \alpha M \rfloor$ because $|\mathcal{A}|$ is an integer. Write $\phi(t) = \mathbf{1}\{\#\{\pi : t_\pi \ge t_{\mathrm{id}}\} \le \alpha M\}$ for a vector t indexed by S_C, so that the rejection indicator is $\phi\big((T(\pi))_\pi\big)$. Because $\pi \mapsto \pi\sigma$ is a bijection of S_C, the vector $(T(\pi\sigma))_\pi$ carries $T(\sigma)$ in its identity coordinate and the same multiset of values, so $\phi\big((T(\pi\sigma))_\pi\big) = \mathbf{1}\{N(\sigma) \le \alpha M\}$, and summing over σ gives $\sum_{\sigma \in S_C} \phi\big((T(\pi\sigma))_\pi\big) = |\mathcal{A}|$. Taking expectations conditional on $\mathcal{Z}$ and applying exchangeability to each term,

$$M \Pr(\hat{p} \le \alpha \mid \mathcal{Z}) = \mathbb{E}\big[\, |\mathcal{A}| \,\big|\, \mathcal{Z}\big] \le \lfloor \alpha M \rfloor,$$

so $\Pr(\hat{p} \le \alpha \mid \mathcal{Z}) \le \lfloor \alpha M \rfloor / M \le \alpha$. Averaging over $\mathcal{Z}$ preserves the bound.

Two cases sharpen it. When the M values are almost surely distinct, N takes each of $1, \dots, M$ exactly once, so $|\mathcal{A}| = \lfloor \alpha M \rfloor$ and the bound holds with equality. When they fall into $M/\mathfrak{m}$ groups of exactly $\mathfrak{m}$ equal values, distinct across groups, which is what Proposition 3(ii) gives with $\mathfrak{m} = 2$ for the two sided Wald statistic, N takes each of $\mathfrak{m}, 2\mathfrak{m}, \dots, M$ with multiplicity $\mathfrak{m}$, so $|\mathcal{A}| = \mathfrak{m}\lfloor \alpha M/\mathfrak{m} \rfloor$ and $\Pr(\hat{p} \le \alpha) = \mathfrak{m}\lfloor \alpha M/\mathfrak{m} \rfloor / M$. In either case the randomized version rejects with probability one on $\{\hat{p} \le \alpha_-\}$, whose probability is α_-, and with probability $(\alpha - \alpha_-)/(\alpha_+ - \alpha_-)$ on $\{\hat{p} = \alpha_+\}$, whose probability is $\alpha_+ - \alpha_-$, so its size is α.

For the Monte Carlo reference set of Definition 2, replace S_C by $\mathcal{R} = \{\mathrm{id}, \pi_2, \dots, \pi_B\}$. Because $\pi_2, \dots, \pi_B$ are i.i.d. uniform on S_C and drawn independently of the data, the vector $(T(\pi))_{\pi \in \mathcal{R}}$ is exchangeable conditional on $\mathcal{Z}$ under Assumption 4, and the rank argument above gives $\Pr(\text{reject}) \le \alpha$ with equality at $\mathfrak{m}\lfloor \alpha B/\mathfrak{m} \rfloor / B$. □

Proof of Proposition 3. (i) Every $\pi \in \mathrm{Stab}(\delta)$ satisfies $\delta_\pi = \delta$ and hence $T(\pi) = T(\mathrm{id})$, so the numerator of $\hat{p}$ is at least $|\mathrm{Stab}(\delta)|$ on the exhaustive reference set, and $\mathrm{Stab}(\delta)$ is the direct product of the symmetric groups on the tied groups of d, of order the product of their factorials. On the Monte Carlo reference set the identity alone gives $\hat{p} \ge 1/B$.

(ii) Because the multiset $\{\delta_c\}$ is symmetric about zero, pick an involution $r \in S_C$ matching each c with $\delta_c = v > 0$ to some c' with $\delta_{c'} = -v$ and fixing every c with $\delta_c = 0$; then $\delta_{r(c)} = -\delta_c$ for all c, and $r \ne \mathrm{id}$ because δ is not identically zero. With equally spaced d we may write $\delta_c = s\,(c - \frac{C+1}{2})$ for a scale $s > 0$ and take $r(c) = C + 1 - c$. Precomposition negates the label vector: $(\delta_{r\circ\pi})_c = \delta_{r(\pi(c))} = -\delta_{\pi(c)}$, so $\delta_{r\circ\pi} = -\delta_\pi$. Evenness of $\mathcal{T}$ then gives $T(r \circ \pi) = T(\pi)$ for every π. The map $\pi \mapsto r \circ \pi$ is left translation by an involution other than the identity, hence a fixed point free involution on S_C, so the M values fall into $M/2$ equal pairs, and $r \circ \mathrm{Stab}(\delta)$ is disjoint from $\mathrm{Stab}(\delta)$ because $r \notin \mathrm{Stab}(\delta)$. The numerator of $\hat{p}$ is therefore at least $2|\mathrm{Stab}(\delta)|$.

For the Wald statistic, $\hat{\gamma}(-\delta) = -\hat{\gamma}(\delta)$ by Lemma 2, each cluster score is odd in δ, and the CR1 covariance is quadratic in the cluster scores, so $\hat{V}_\gamma(-\delta) = \hat{V}_\gamma(\delta)$ and the quadratic form is unchanged.

(iii) Rejection at level α requires some attainable $\hat{p} \le \alpha$, hence $2|\mathrm{Stab}(\delta)|/C! \le \alpha$ in the even case and $|\mathrm{Stab}(\delta)|/C! \le \alpha$ otherwise, which reduce to $2/C! \le \alpha$ and $1/C! \le \alpha$ when d has no ties. □

Proof of Theorem 2. For the first claim, replace S_C in the proof of Theorem 1 by the finite group $\mathcal{E} = \{-1, +1\}^G$ acting by coordinatewise sign change. That proof uses only finiteness of the group and invariance of the data law under right translation by each of its elements, both of which the sign symmetry hypothesis supplies. $\mathcal{E}$ is abelian, so the distinction between left and right translation that the proof of Theorem 1 turns on is vacuous here.

For the second claim, let u_{gc} denote the cluster-g category-c errors under (4.2), taken before the two way within transformation, and build the infeasible contribution from them,

$$\zeta_g^0 = \frac{\big(S^{-1} \sum_c \delta_c \tilde{X}_g' u_{gc}\big)_P}{\sum_c \delta_c^2}.$$

Using the raw rather than the within transformed errors matters here and costs nothing. It matters because the two way within projection subtracts period means, which pool across clusters, so contributions formed from the within transformed errors are dependent across g. It costs nothing because $\tilde{X}'\Pi = \tilde{X}'$, so the aggregate is unchanged and $\sum_g \zeta_g^0 = \hat{\gamma}_P - \gamma_P$ still holds. Each ζ_g^0 is then a fixed linear functional of cluster g's errors alone, so independence across g together with symmetry of each summand about zero gives $(\epsilon_g \zeta_g^0)_g \overset{d}{=} (\zeta_g^0)_g$ for every $\epsilon \in \mathcal{E}$.

For the third claim, write z_g^r for the cluster-g score of the γ block from the fit with (4.2) imposed, so the feasible contribution is $\zeta_g = (S^{-1} z_g^r)_P / \sum_c \delta_c^2$. It replaces the errors by the restricted residuals, which subtracts a term $S^{-1} S_g [\sum_h (\cdot)_h]$. The factor $\|S_g\|/\|S\|$ is $O(G^{-1})$ and it multiplies a sum of G cluster contributions, which is $O_p(G^{1/2})$ times a single one; relative to ζ_g itself the discrepancy is therefore $O_p(G^{-1/2})$.

Asymptotic level α does not follow from an approximate symmetry argument over $\mathcal{E}$. The results of Canay et al. (2017) hold for a randomization group whose cardinality is fixed in the sample size, and $|\mathcal{E}| = 2^G$ grows with G. We argue it instead as a statement about the conditional law of the randomization distribution. Assume (W1) the cluster blocks $(\tilde{X}_g, \{u_{gc}\}_c)$ are independent across g with $\mathbb{E}[z_g^0] = 0$ under (4.2) and uniformly bounded moments of order strictly greater than four, where $z_g^0 = \sum_c \delta_c \tilde{X}_g' u_{gc}$; (W2) $\max_g \|S_g\|/\|S\| \to 0$ and $G^{-1} S \to \bar{S} \succ 0$; and (W3) $G^{-1} \sum_g \mathbb{E}[z_g^0 z_g^{0\prime}] \to \Sigma_z \succ 0$ on the block P. The sign change distribution is the null imposed Rademacher wild cluster bootstrap of Cameron et al. (2008) in score form. Conditional on the data, $G^{-1/2} \sum_g \epsilon_g z_g^r$ is a sum of G independent mean zero terms with conditional variance $G^{-1} \sum_g z_g^r z_g^{r\prime}$, which converges in probability to Σ_z under (W1) to (W3), and (W1) with (W2) gives the conditional Lyapunov condition. The conditional bootstrap central limit theorem of Djogbenou et al. (2019) then gives the randomization distribution of the statistic the same χ^2_m limit, conditionally in probability, that the statistic itself has under (4.2). The randomization p value is therefore asymptotically uniform. □

Proof of Corollary 3. Under the alternative $\gamma = \kappa k$, Lemma 2 gives $\hat{\gamma} = \sum_c \delta_c b_c / \Sigma_\delta^2$ with $\mathbb{E}[b_c] = (\beta_\ell)_{\ell \in L} + \delta_c \kappa k$, and with homoskedastic categories independent across c, $\mathrm{Var}(b_c) = \sigma^2 S^{-1}$. Since $\sum_c \delta_c = 0$, $\hat{\gamma}$ has mean κk and covariance $(\sigma^2/\Sigma_\delta^2) S^{-1}$, so on the post treatment block $V = \mathrm{Var}(\hat{\gamma}_P) = (\sigma^2/\Sigma_\delta^2)[S^{-1}]_{PP}$ and, by the definition of I_k in Section 8, $\kappa^2 k_P' V^{-1} k_P = \kappa^2 \Sigma_\delta^2 I_k / \sigma^2$.

The first expression in (8.2) is the minimum detectable response of the infeasible Wald statistic that uses V in place of an estimate. That statistic is noncentral chi square with m degrees of freedom and noncentrality $\lambda = \kappa^2 \Sigma_\delta^2 I_k / \sigma^2$; setting $\lambda = \lambda^*(\alpha, \beta, m)$ and solving

for κ gives it. The feasible CR1 Wald statistic does not have that law. The identity in the proof of Proposition 2 uses no distributional assumption and holds under the alternative with $\bar{\eta}_P$ replaced by $\hat{\gamma}_P/G$, so under Assumptions 2 and 3 the feasible statistic is ω_2 times a noncentral Hotelling T^2 with dimension m and $G-1$ degrees of freedom. Equation (8.2) therefore understates the minimum detectable response of the feasible test twice, through the critical value, as Section 8 records, and through the distribution under the alternative.

For the directed statistic, t is standard normal under (4.2) and normal with unit variance and mean $\kappa\sqrt{\Sigma_\delta^2 I_k}/\sigma$ under the alternative, so power $1-\beta$ at level α requires that mean to equal $z_{1-\alpha}+z_\beta$, which gives the second expression. □

Proof of Theorem 3. Two configurations $(b_c)_c$ with the same contrast vector $(b_c-\bar{b})_c$ differ by a shift common to all categories, so the $C-1$ contrasts are a maximal invariant and an invariant test is a function of them. The maximal invariant is the full contrast vector, of dimension $(C-1)q$, and not $\hat{\gamma}_P$. Decompose the contrast space orthogonally in the category index into the direction δ and its complement. The δ-component is $\hat{\gamma} = \sum_c \delta_c b_c / \sum_c \delta_c^2$, with mean κk. The remaining $C-2$ contrasts have mean zero for every κ, so they are ancillary for κ, and because Σ does not vary with c they are uncorrelated with $\hat{\gamma}$ and, being jointly Gaussian with it, independent of it. Write A for that block.

Conditional on $A=a$, the law of $\hat{\gamma}_P$ is $N(\kappa k_P, V)$ and does not depend on a. Its density in κ is proportional to $\exp\{\kappa k_P' V^{-1}\hat{\gamma}_P - \frac{1}{2}\kappa^2 k_P' V^{-1} k_P\}$ relative to the $\kappa=0$ density, a one parameter exponential family with natural parameter κ and natural sufficient statistic $U = k_P' V^{-1}\hat{\gamma}_P$, which has monotone likelihood ratio in κ. By the Karlin–Rubin theorem (Lehmann and Romano 2005) the conditional test rejecting for large U is uniformly most powerful at every conditional level, and at conditional level u its power is $\Psi(u) = \Phi(\mu + \Phi^{-1}(u))$ with $\mu = \kappa\sqrt{k_P' V^{-1} k_P} > 0$.

An invariant test may still randomize its level over the ancillary block, taking conditional level $\alpha(A)$ with $\mathbb{E}[\alpha(A)] = \alpha$, and ruling that out is the remaining step. Differentiating, $\Psi'(u) = \exp\{-\mu^2/2 - \mu\Phi^{-1}(u)\}$, which is strictly decreasing in u, so Ψ is strictly concave on $(0,1)$ and Jensen's inequality gives $\mathbb{E}[\Psi(\alpha(A))] \le \Psi(\alpha)$, with equality if and only if $\alpha(\cdot)$ is almost surely constant. The unconditional power of an invariant level-α test is therefore at most $\Psi(\alpha)$, attained by $\alpha(\cdot)\equiv\alpha$. Standardizing U by $\sqrt{k_P' V^{-1} k_P}$ gives $t \sim N(\kappa\sqrt{k_P' V^{-1} k_P}, 1)$, from which the critical value and the stated power function follow. □

S2 Additional simulation evidence

The exact null law in the ideal experiment. Assumption 3 is not satisfied by the panel design, so Proposition 2 is checked separately in an experiment built to satisfy it: cluster contributions drawn i.i.d. $N(0,\Sigma)$ with $m=13$, Σ redrawn for each of three seeds, and 60,000 replications in total at each G. The identity $W = \omega_2 T^2$ holds to a maximum relative error of 4.0×10^{-14} at $G=20$ and 1.3×10^{-15} at $G=100$. Referring the statistic to the F law of (5.2) at the nominal G gives null rejection rates of 0.0495, 0.0489, 0.0502 and 0.0499 at $G=20$, 30, 51 and 100, quoted to four decimals here because the claim is that they are indistinguishable from the nominal 0.05 rather than merely close to it. Kolmogorov–Smirnov tests of the resulting p values against the uniform return $p=0.81$, 0.02, 0.06 and 0.77 at those four G, so the pooled $G=30$ draws reject uniformity at the 5% level. The closed form chi square size implied by (5.2) matches the simulated one to within 0.003 at every G. Source: `anatomy_identity.csv`.

Accuracy of the substitution of G_{eff}. Evaluating (5.2) at G_{eff} from Definition 1 predicts a null rejection rate of 0.378 against the observed 0.389 at $G = 51$, $C = 5$, $m = 13$. Across the twelve (G, C) cells the mean absolute prediction error is 2.37 percentage points using G_{eff} and 13.13 percentage points using G. In this design $G_{\text{eff}} = 36.42$, or 71.4% of G, and it is a design object rather than a random one: its standard deviation across replications is 10^{-13}. The Hotelling calibration evaluated at G_{eff} tracks the nominal level only in the middle of the grid. It rejects at 0.000 in all three cells at $G = 20$, at 0.022, 0.020 and 0.021 at $G = 30$, at 0.047, 0.045 and 0.045 at $G = 51$, and at 0.061, 0.065 and 0.055 at $G = 100$, so the substitution is conservative at twenty clusters and mildly liberal at a hundred.

When exchangeability fails. A departure from Assumption 4 is priced in a Gaussian limit experiment rather than in the panel: one $N(0, \omega_c)$ draw per category, a single restriction, the same randomization over the $C!$ label assignments, a covariance estimate that treats the categories as homoskedastic, and 20,000 replications. With the inflated variance on the highest exposure category, the label permutation test using the Wald statistic rejects at 0.050, 0.064, 0.077, 0.090 and 0.102 at $C = 5$, and at 0.048, 0.062, 0.079, 0.100 and 0.125 at $C = 8$, for variance ratios 1, 2, 4, 9 and 25. Placing the inflated variance on the lowest exposure category instead moves those rates by at most 0.009. The distortion is monotone in the ratio, and its largest value in the grid stays below the chi square rate of 0.356 measured at $G = 51$, $C = 8$, $m = 13$. DiCiccio and Romano (2017) show for a regression coefficient that studentizing the statistic makes the permutation test exact under independence and asymptotically valid under heteroskedasticity, so a studentized category level statistic is the natural repair for this departure; we have not derived its behaviour under Assumption 4.

Two further experiments show that the rates above overstate what the departure costs the procedure we run. The homoskedastic covariance estimate of the limit experiment is not the one the panel uses. The CR1 meat $\sum_g \eta^f_{gP}\eta^{f\prime}_{gP}$ is quadratic in the realized residuals, which enter it category by category weighted by $\delta_{\pi(c)}$, so its variance estimate already carries a separate contribution per category and already moves with π. Supplying the limit experiment with a variance estimate of the same kind, $\hat{\omega}_c = \omega_c \chi^2_\nu / \nu$ drawn independently across c, reverses the sign of the distortion: with the inflated variance on the highest exposure category the studentized Wald statistic rejects at 0.050, 0.039, 0.028, 0.015 and 0.006 at $C = 5$ and at 0.048, 0.040, 0.030, 0.018 and 0.005 at $C = 8$ as ν grows without bound, over the same grid. Studentizing leaves the exchangeable case at nominal size and makes the departure conservative rather than liberal.

The panel behaves the same way. Running the same departure through the design of this section with the CR1 statistic itself, drawing category error standard deviations that deliver median realized residual variance ratios of 2.77, 3.07, 5.09, 11.22 and 32.90 over 2,000 replications per cell, the two sided label permutation test rejects at 0.057, 0.045, 0.032, 0.020 and 0.009 at $C = 5$ and at 0.054, 0.048, 0.038, 0.027 and 0.009 at $C = 8$, with a Monte Carlo standard error of at most 0.005 across the forty randomization cells. The directed version rejects at 0.051, 0.040, 0.032, 0.023 and 0.010 and at 0.050, 0.046, 0.041, 0.033 and 0.016. No cell exceeds the nominal level, and the rates track the studentized limit experiment at $\nu = 10$, which gives 0.051, 0.042, 0.031, 0.020 and 0.009 at $C = 5$. Against a category variance ratio of 33 the cost of the departure is power and not level. Source: `studentized_check.csv`.

In the limit experiment with a homoskedastic covariance estimate the sign of the distortion depends on how the variance spread lines up with the exposure index. In the implemen-

tation used for Tables 2 and 3, which recomputes the CR1 covariance for every permuted label vector, it does not. Moving the inflated variance to the lowest exposure category instead of the highest gives 0.057, 0.039, 0.028, 0.024 and 0.015 at $C = 5$ and 0.054, 0.043, 0.033, 0.031 and 0.011 at $C = 8$, so the panel rejection rate falls with the variance ratio under either placement. A separate $C = 6$ design with category variances decreasing in exposure moves the same way, from 0.047 at variance ratio 1 to 0.021 at ratio 25 over 2,000 replications. Sources: `exchangeability_failure.csv` and `studentized_check.csv`, and `a2_size_heteroskedastic.csv` under `results/inference/`.

The price of label randomization. Relative to a normal benchmark at level 0.05 and power 0.80, the label permutation test requires the standardized signal inflated by a factor of 1.892 at $C = 4$, 1.352 at $C = 5$, 1.235 at $C = 6$, 1.149 at $C = 8$ and 1.098 at $C = 10$, so the price falls as C grows. The factor is a large sample calculation and it holds up only at the larger C. The minimum detectable response it implies is 0.231 at $C = 8$ against 0.243 in simulation, and 0.327 at $C = 5$ against 0.477, where the simulated value is 46% larger, at the smallest C for which Proposition 3(iii) makes the two sided test usable. Sources: `perm_power_ceiling.csv` and `perm_mde_vs_sim.csv`.

Exposure spread, panel length and the pre window. Three further blocks vary the design rather than the procedure, at $G = 51$ and $C = 5$. Compressing the exposure indices from $[0.1, 1]$ to $[0.4, 0.6]$ cuts Σ_δ^2 from 0.506 to 0.025 and raises the closed form minimum detectable response of the Wald test from 0.363 to 1.639, a factor of 4.5, with the directed test moving from 0.242 to 1.854. Halving the panel to $T = 60$ moves the Wald figure by under 2%, from 0.363 to 0.370. Replacing the six period pre window by the single omitted period $\ell = -1$ raises I_k from 23.6 to 448 and cuts the closed form directed minimum detectable response to 0.050, against 0.081 for the simulated directed cluster sign change test; the chi square null rejection rate in that block is 0.405 over 600 replications, so narrowing the pre window buys power without repairing the calibration. Sources: `mde_table.csv` and `reference_block_table.csv`.

Robustness. Four designs drop the features of the block above that favour the procedures we propose, each at $G = 51$, $C = 5$ and 1,000 replications, against a Monte Carlo standard error of 0.007 at a rate of 0.05. The first gives every cluster five units and a cluster by period shock with standard deviation 0.6, so that the clustering carries dependence rather than labelling singletons. The second adds unequal cluster sizes, spread geometrically from two units to sixteen at the same total, which is the configuration MacKinnon and Webb (2017) identify as the wild bootstrap's hardest case. The third replaces the independent errors by an AR(1) in the period index with coefficient 0.7. The fourth replaces the Gaussian errors by standardized t_5 draws, so that only the tails change.

The label permutation test holds its size in all four, rejecting at 0.050, 0.058, 0.059 and 0.056 against 0.046 in the baseline design. Nothing in Theorem 1 leads one to expect otherwise, since Assumption 4 restricts the category direction alone and is untouched by cluster size, serial dependence or tails, but the measurement is what the theorem is worth if the theorem is right. The chi square calibration over rejects in all four, between 0.307 and 0.372 against 0.394 in the baseline design.

Two procedures degrade. The restricted wild cluster bootstrap moves from 0.048 to 0.066 and 0.065 in the two clustered designs. The directed cluster sign change test moves from 0.059

to 0.094 and 0.086, which is five Monte Carlo standard errors above nominal and is the reason the recommendation in Section 1 is qualified. Its two sided counterpart is far less affected, at 0.057 and 0.063 against 0.050. The pattern is consistent with Theorem 2: the finite sample statement covers the infeasible statistic, the feasible one differs by $O_p(G^{-1/2})$, and within cluster dependence is what makes that gap bite. A one sided statistic divided by a noisier estimated standard deviation is where it shows first.

A bias reduced variance does not repair the calibration. The literature on few clusters offers a better variance estimator than the CR1 used here: the bias reduced linearization of Bell and McCaffrey (2002), which premultiplies each cluster's residuals by $(I - H_{gg})^{-1/2}$ before forming the meat. Applying it changes almost nothing. Across the five designs the chi square rejection rate moves from 0.394, 0.307, 0.361, 0.358 and 0.372 under CR1 to 0.390, 0.305, 0.357, 0.353 and 0.372 under CR2, every difference inside one Monte Carlo standard error, and no cluster in any design has leverage high enough for the adjustment to be undefined. In the two applications it moves the crime chi square p value from 0.025 to 0.026 and the two health values from 0.730 and 0.391 to 0.736 and 0.435. The reason is that the two corrections address different things. Bell and McCaffrey (2002) correct the downward bias of the meat matrix, which is an $O(1/n_g)$ effect and is small when each cluster contributes many periods, while the distortion this paper is about is the substitution of a chi square for a distribution whose denominator has $G - m$ degrees of freedom, and that survives any reweighting of the residuals. The exact law of Proposition 2 is stated for CR1 and does not transfer, so we do not report a Hotelling calibration of the CR2 statistic. Source: `robust.csv`.

S3 Additional application evidence

The second panel is a state by year by condition panel of BRFSS crude prevalence covering 51 jurisdictions from 2011 to 2024, clustered on state. The treatment is a state's adoption of the ACA Medicaid expansion, with the cohort set to the calendar year of the coverage effective date when coverage began on or before July 1 and to the following year otherwise; ten states do not expand through 2024. Seven conditions carry the gradient, and the exposure index grades them by whether a clinical encounter is required for the respondent to report the condition: diagnosed diabetes 1.00, diagnosed high cholesterol 1.00, diagnosed hypertension 0.90, diagnosed depression 0.80, self reported obesity 0.05, current smoking 0.00 and no exercise in the past month 0.00. Three further series in the panel, diabetes mortality, insurance coverage and a past year checkup, are not used in the gradient.

Hypertension and high cholesterol are asked only in odd years, so Assumption 2 fails on the seven category stack. That failure costs the closed form of Lemma 2 and not the conclusion of Theorem 1, which needs Assumption 4 in its place. What blocks exactness on the full set is that the cell pattern is a function of the category label, so Assumption 4 fails as well. The cluster level procedures of Section 7 use neither assumption and could be run on the unbalanced full set; we do not report that specification. We report two balanced sub designs: the five annually asked categories on all 709 state year cells, and all seven categories on the 353 odd year cells. Under either equivariant normalization no category is dropped, so $C = 5$ and $C = 7$, with $G_{\text{eff}} = 36.0$ and 32.8, $\Sigma^2_\delta = 0.958$ and 1.444, and support for $m = 2$ restrictions against the $m = 7$ tested. Neither sub design rejects (4.2) under the triad normalization, at chi square $p = 0.730$ and 0.391 and label permutation $p = 0.850$ and 0.854 over all $C!$ assignments, nor

Table 1: Crime application: p values for $H_0 : \gamma_P = 0$ across nine designs and three loading normalizations

Design	Normalization	C	G_{eff}	χ^2	Hotelling	Perm.	Bootstrap	Sign
Baseline	reference	6	45.9	0.018	0.097	0.096	0.075	0.083
	naive			0.141	0.304	0.396	0.277	0.311
	triad			0.025	0.115	0.115	0.092	0.104
Pre 2020	reference	6	33.2	0.367	0.604	0.782	0.630	0.593
	naive			0.596	0.769	0.911	0.773	0.766
	triad			0.422	0.647	0.813	0.670	0.639
Drop rape	reference	5	45.9	0.016	0.089	0.075	0.063	0.048
	naive			0.008	0.062	0.067	0.040	0.052
	triad			0.018	0.096	0.083	0.066	0.051
Drop rape, larceny	reference	4	45.9	0.107	0.258	0.667	0.218	0.209
	naive			0.139	0.300	0.667	0.254	0.303
	triad			0.106	0.257	0.667	0.220	0.203
Cohort +1	reference	6	45.7	0.017	0.094	0.117	0.079	0.078
	naive			0.120	0.277	0.371	0.258	0.299
	triad			0.023	0.110	0.122	0.097	0.099
Threshold 0.8	reference	6	38.0	0.182	0.394	0.243	0.411	0.367
	naive			0.000	0.007	0.011	0.002	0.003
	triad			0.204	0.419	0.268	0.431	0.399
Core agency types	reference	6	46.9	0.012	0.074	0.032	0.062	0.062
	naive			0.164	0.328	0.389	0.306	0.359
	triad			0.018	0.093	0.051	0.083	0.091
Drop never treated	reference	6	45.3	0.017	0.094	0.172	0.093	0.094
	naive			0.079	0.220	0.338	0.216	0.245
	triad			0.026	0.119	0.207	0.121	0.131
Core types, pre 2020	reference	6	35.8	0.320	0.547	0.793	0.566	0.554
	naive			0.537	0.717	0.926	0.728	0.717
	triad			0.367	0.587	0.819	0.601	0.602

Note: p values for the joint null $H_0 : \gamma_P = 0$ of (4.2) in the pooled gradient regression (4.1), estimated by OLS with the CR1 covariance clustered on state. Every design has $G = 51$ state clusters except drop never treated, which has $G = 47$, and every design tests $m = 9$ post treatment restrictions. χ^2 is the χ^2_9 calibration; Hotelling is (5.2) evaluated at G_{eff}; Perm. is the label permutation test of Definition 2 enumerating all $C!$ assignments of δ, whose realized resolution floor is $1/720$ at $C = 6$, $1/120$ at $C = 5$ and $1/24$ at $C = 4$; Bootstrap is the restricted wild cluster bootstrap and Sign the cluster sign change test, each with 1,999 draws. Naive sets every loading to one; triad is the S_C-equivariant estimator (6.1); adjusted fixes the reference category at the smallest d_c and is not covered by Theorem 1. The entry shown as 0.000 is 2.1×10^{-5}. Sources: `results/empirics/inference_gradient.csv` for the p values and `results/empirics/app_floors_injection.csv` for the realized floors.

under the naive normalization, at chi square 0.295 and 0.967 and label permutation 0.867 and 0.964.

The health panel is also where the one factor premise behind (6.1) is least credible. By the quantities of Section 6 the common factor carries 0.121 and 0.065 of the median category's demeaned variance in the two sub designs, against 0.397 in the crime panel, and the single triads disagree about ρ_c by factors of 2.39 and 8.59. The nonrejections above are therefore weak evidence in two separate senses. The design supports $m = 2$ restrictions and tests $m = 7$, and the normalization the equivariance of Theorem 1 runs through is estimating loadings on a factor that carries little of the variation. Neither point bears on the crime design, where the factor is the strongest of the three.

Every rejection anywhere in the health panels comes from the one normalization Theorem 1 does not cover, and none survives the correct reference set. The reference category rule fixes its reference at the position of the smallest d_c, which is not S_C-equivariant, and its loading guard drops diagnosed diabetes in the annual sub design and high cholesterol and no exercise in the odd year sub design. Under it the odd year sub design rejects at $p = 0.033$ on the level outcome and under every calibration on the log outcome. Where the guard fires, permuting only the survivors conditions on a multiset that is not ancillary: Assumption 4 makes the reference set the $C_0!/(C_0 - C)!$ ordered injections rather than the $C!$ permutations of the survivors, and recomputing against it moves the odd year value from 0.033 to 0.062. Under the two equivariant normalizations nothing rejects on either outcome. Given Section 10.1 none of this is evidence about Medicaid expansion, and we report it because a normalization that is not equivariant is the failure mode Theorem 1 is designed to exclude and this is what it looks like in data.

The chi square rejection is joint with the normalization. Under the naive normalization the chi square test does not reject at all, at $p = 0.141$, and the label permutation returns $p = 0.396$; under the reference category normalization the chi square test gives $p = 0.018$ and the label permutation $p = 0.096$. Table 1 reports all three normalizations for each of the nine crime designs, of which the design described above is one. The artifact that correct calibration overturns is therefore an artifact of the chi square reference distribution applied to a normalized outcome; under the naive normalization there is nothing to overturn.

A pre period placebo in the same design behaves differently. Under the triad normalization its chi square p value is 6×10^{-6} and its label permutation p value over the same 720 assignments is 0.0167, the twelfth largest of the 720 orbit values. The exposure indices are not symmetric about zero and the realized orbit multiplicity is one, so Proposition 3(ii) does not apply and the floor is $1/720$ rather than $2/720$. Under the naive normalization the same permutation test returns $p = 0.153$, so the pre period evidence, like the post period evidence, is conditional on the normalization.

S4 Additional identification and normalisation evidence

The three designs of Section 10 differ sharply on both. In the crime panel the factor carries 0.397 of the median category's demeaned variance, the largest eigenvalue of the cross category correlation matrix carries 0.522 of its trace against the $1/C = 0.167$ that no factor at all would give, and single triads disagree about ρ_c by a factor of 1.23. In the annual and odd year health sub designs the median shares are 0.121 and 0.065, the trace shares are 0.269 and 0.241 where no factor would give 0.200 and 0.143, and the single triads disagree by factors of 2.39 and 8.59.

One category in the odd year sub design has a share of 0.004. The premise is a serviceable approximation in the crime design and is not one in the health designs, where (6.1) averages quantities that do not agree with each other.

These are descriptive quantities and we attach no p value to them. A bootstrap reference fitted under a one factor null is not calibrated for either: on data generated from a true one factor structure at the dimensions of the three designs, with and without serial correlation in the factor and in the idiosyncratic errors, a circular block bootstrap reference rejects between 0.090 and 0.995 on the eigenvalue ratio and between 0.090 and 0.178 on the triad dispersion, against a nominal 0.05 over 400 replications per cell. Fitting the null to the data biases both references, and we report the statistics rather than tests built on them. Sources: `onefactor_diagnostic.csv` and `onefactor_calibration.csv`.

What a failure of the premise costs is worth stating precisely. Lemma 3 is algebraic and holds whatever generates the data, and one factor is not among the hypotheses of Theorem 1, which are Assumption 4 and order invariance. What the structure supplies is the reason to expect Assumption 4 to hold after normalizing, since dividing by an estimated loading equalizes the factor component of the arrays only when there is a common factor to equalize. Where the premise fails the conclusion of Theorem 1 is untouched and the case for its hypothesis is weaker, which is a reason to report the cluster level procedures of Section 7, since they use neither the normalization nor Assumption 4.

A second reading deserves an answer. Mikhaeil and Harshaw (2025) describe the same collapse and treat it as correct behaviour, on the ground that an explicit conditional extrapolation assumption licenses inferring small post treatment violations from small pre treatment ones. That defence is sound where the assumption holds. A recording response is precisely a violation of it that passes every pre test by construction, so the assumption cannot be checked in the direction that matters here.

For the smoothness class the corresponding statement is weaker and older. That a researcher cannot calibrate M from the data and retain honest coverage is the nonadaptivity result of Low (1997) and Cai and Low (2004), stated for econometric practice by Armstrong and Kolesár (2018) and Armstrong and Kolesár (2020), who write that "to maintain honesty over the whole function class, a researcher must choose M a priori, rather than attempting to use a data driven method." Kolesár and Rothe (2018) put the asymmetry sharply: the data can bound the smoothness constant from below but never from above, and an honest interval needs an upper bound. Freeman (2026) restates the principle in a difference-in-differences framing. Nothing in this paper improves on any of that. What is new is the connection: $\Delta^{\mathrm{SD}}(M)$ is convex and centrosymmetric, so the nonadaptivity result applies to it directly, and Rambachan and Roth (2023) do not cite it. Their reference list contains Low (1995) for the computation of fixed length intervals and Armstrong and Kolesár (2018, 2020) for their optimality, but neither Low (1997) nor Cai and Low (2004).

S5 Related work and the applied literature

Related work

Our exact null law is a statement about the cluster robust Wald statistic with G clusters, which places it in the few cluster inference literature. Cameron et al. (2008) propose the wild cluster bootstrap with the null imposed and document its size improvement over the cluster

robust t; MacKinnon and Webb (2017) characterize when it fails. Neither paper treats the joint m restriction case in which m is comparable to G, and that is precisely where we find cross outcome falsification tests. Carter et al. (2017) show that cluster heterogeneity reduces the number of clusters that the asymptotics effectively deliver, and define the effective cluster count we use in Section 5. Canay et al. (2017) give asymptotic validity for a randomization test built on a group of fixed cardinality when the invariance holds only in the limit. The label permutation group of Section 6 has cardinality $C!$, which does not vary with G, so their framework covers designs in which the exchangeability of Assumption 4 holds for the category statistics only as G grows. The finite sample argument itself is the classical group invariance construction, for which we follow Lehmann and Romano (2005).

We draw on four further results directly. MacKinnon et al. (2023) set out current practice for cluster robust inference, and Section 7 takes the restricted wild cluster bootstrap in the form given there. Ibragimov and Müller (2010) obtain a t-test on cluster level estimates whose level is controlled with a small fixed number of heterogeneous clusters and no appeal to $G \to \infty$; that construction and the label permutation test of Section 6 differ in what is treated as the repeated unit, clusters there and outcome categories here. Young (2019) recomputes p values by randomization inference for a sample of published experimental papers in economics and reports that a substantial share of the rejections at conventional levels do not survive, which is the direction of the gap between the chi square and the permutation p value in Section 10. Djogbenou et al. (2019) prove a central limit theorem for the wild cluster bootstrap distribution conditional on the data, and that theorem supplies the asymptotic level of the cluster sign change test of Section 7, which is the null imposed Rademacher wild cluster bootstrap written in score form.

We are not the first to correct both the variance estimator and the reference distribution for statistics of this kind, and a small sample cluster robust literature does both. Bester et al. (2011) hold the number of clusters fixed and let cluster size grow, and obtain limiting t and F distributions in which G plays the role of the sample size; the $F(m, G-m)$ shape of (5.2) is theirs as a large cluster approximation. Bell and McCaffrey (2002) remove the downward bias of the cluster robust variance estimator under a working model and pair the correction with Satterthwaite degrees of freedom, and Pan and Wall (2002) do the same for generalized estimating equations. Imbens and Kolesár (2016) give degrees of freedom adjustments for the heteroskedasticity robust case. For a hypothesis with $m > 1$ constraints, Tipton and Pustejovsky (2015) refer the statistic to an approximate Hotelling T^2 whose denominator degrees of freedom are matched from the data, and Pustejovsky and Tipton (2018) extend that test to models with arbitrary sets of fixed effects and report Type I error close to the nominal level across the designs they simulate.

Proposition 2 does not introduce that reference distribution. It states an algebraic identity between the CR1 Wald statistic and a Hotelling T^2, and, under Assumption 3, an exact $F(m, G-m)$ law for the joint null that the post treatment coefficients do not vary with exposure, holding at every G and m with $G - m \geq 1$ rather than at denominator degrees of freedom matched from the data, and rather than in the large cluster limit of Bester et al. (2011). What we add is the finite sample statement and the explicit constant ω_2, rather than the family itself. Corollary 2 is the second difference. It gives the rate $m = o(G^{2/3})$ at which the chi square calibration stays valid, the limit the size converges to when $m^{3/2}/G$ tends to a positive constant, and the convergence of the size to one when m/G does. Calhoun (2011) derives the corresponding correction for the regression F test when the number of restrictions

grows with the number of observations rather than with the number of clusters, and reports the same direction of failure for the uncorrected test.

The design that motivates the problem is the use of graded outcome categories to detect a treatment induced change in how outcomes are recorded. Weisburd et al. (2024) run this contrast across crime categories. Hausman et al. (1998) formalize outcome misclassification that depends on treatment status in a cross sectional discrete response setting, and Gutknecht and Liu (2024) study misclassified outcomes in a changes in changes design, and Denteh and Kédagni (2022) study a misclassified treatment indicator in difference-in-differences, a distinct channel: their designs maintain parallel trends and ask what misclassification does to the estimand, whereas an outcome recording response is itself a parallel trends violation. Freyaldenhoven et al. (2019) use an auxiliary series that loads on the same latent confound as the outcome and use it to purge event study coefficients of that confound. The device is close to the normalization that Assumption 4 calls for here, and it is put to a different use: dividing each category outcome by an estimate of its loading on the common factor equalizes those loadings, which is what makes the category arrays exchangeable, and it removes nothing from the coefficient under test. We take the estimand and the identification argument from that literature as given. This paper is about the reference distribution.

Falsification tests are also run as pre trend tests, where Roth (2022) shows that conditioning on passing one distorts the sampling distribution of the effect estimate, and Rambachan and Roth (2023) replace the test with bounds on the post treatment violation. Our own results concern the calibration of a cross outcome test rather than the consequences of conditioning on its outcome, and we note that the two problems compound, since a test with size 0.39 conditions on an event that is mostly noise.

The gap is a live one. Reviewing difference-in-differences designs for pandemic policy, Goodman-Bacon and Marcus (2020) warn that estimates "will be biased if testing improves differently in the treatment and control groups," then recommend event study estimates, which "can reveal many of the biases outlined above." Corollary 1 says that when the policy is what changed the recording, that remedy rejects no more often than its nominal level. Richardson et al. (2025) report a formal pre trend test as evidence that parallel trends holds, and separately concede that their estimated increase in reported intimate partner violence may reflect disclosure rather than incidence, without connecting the two. Baum et al. (2026) evaluate prenatal screening mandates, a treatment whose direct effect is on detection, and validate the design with a flat pre period and a placebo in time test. Callaway and Li (2023) report that they cannot detect violations of parallel trends in pre treatment periods even though those violations are probably present, and attribute the failure to pre tests being underpowered. Corollary 1 says that where the contamination is a recording response the failure is not a matter of power.

Where the threat is taken seriously, the response is already the method we propose here. Harasztosi and Lindner (2019) study a large Hungarian minimum wage increase and validate the design on the pre period, reporting that exposed and unexposed firms "follow a parallel trend before the minimum wage hike" and that the trend "broke exactly in 2001, when the minimum wage was raised." In an appendix they concede that "in the presence of tax evasion, some of the estimated effects may reflect only reporting behavior," and that firms absorbing the shock by declaring previously undeclared earnings would make "the change in reported wages overstate the actual wage change." They settle the question not from the pre period but across outcomes: "if the main response to the minimum wage is simply reporting, it is not clear why firms would adjust their capital stock or raise their prices," and a reporting story

predicts a negative effect on materials that the data do not show. That is a cross outcome falsification test conducted informally, and Corollary 1 explains why the pre period could not have served instead. The rest of this paper supplies the reference distribution such a test needs when the number of clusters is small.

The sharpest illustration is a design that works. Studying upcoding in Medicare Advantage, Geruso and Layton (2020) observe that "because MA risk scores are calculated using diagnosis data from the prior plan year, changes in MA enrollment can plausibly affect reported risk scores via differential coding only with a lag. Thus, contemporaneous penetration acts as a placebo test." Their placebo has power because an institutional accident supplies a window in which the coding response cannot yet have operated, and the identifying content is that window rather than the pre period as such. Corollary 1 is the statement of what is left when no such window exists.

S6 Three findings resolved by replication

Three findings in Section 9 that an earlier draft reported without a cause are resolved by replication. The measured chi square size falls from 0.393 at $C = 3$ to 0.356 at $C = 8$ at $G = 51$, a decline of 2.45 standard errors, where (5.2) predicts almost no movement, and the draft attributed it to the loading normalization. An independent run of the same twelve cells at 2,000 draws gives 0.388, 0.385 and 0.384, flat to within a quarter of a standard error, and removing the normalization entirely gives 0.396, 0.391 and 0.396, also flat. The decline is neither the normalization nor a feature of the design.

The size adjusted power of the two sided label permutation test at $C = 5$ rises to 0.452 at $\kappa = 0.60$ and falls to 0.383 at $\kappa = 0.80$ in Table 3, a difference of 2.4 standard errors. Two candidates separate. The realized signal is strictly monotone: over 600 replications per cell the mean post treatment gradient rises through 0.141, 0.267, 0.374, 0.450 and 0.488 and the median Wald statistic through 24.7, 36.8, 56.1, 70.2 and 79.0 as κ rises from 0.15 to 0.80, so neither the statistic nor the log wedge turns over. Size adjusted power over the same grid runs 0.093, 0.193, 0.335, 0.335 and 0.348, which is a plateau and not a decline. What the permutation test has at $C = 5$ is a ceiling rather than a turning point: the reference set has 120 points, and once the observed statistic is larger than most of the relabelled ones a further increase in κ cannot raise its rank. The large sample local power calculation, which knows nothing of the reference set, is the same statement from the other side, and it is why it overstates simulated power by between 15 and 54 percentage points here.

The Kolmogorov Smirnov test of p value uniformity rejects at $G = 30$ in the experiment built to satisfy Assumption 3. Repeating that test on 200 independent sets of 1,000 draws gives rejection rates of 0.060, 0.045, 0.035 and 0.060 at $G = 20$, 30, 51 and 100, each within one Monte Carlo standard error of 0.015 of the nominal 0.05. The reported rejection is what a correctly sized test does one time in twenty. Source: `anomalies.csv`.

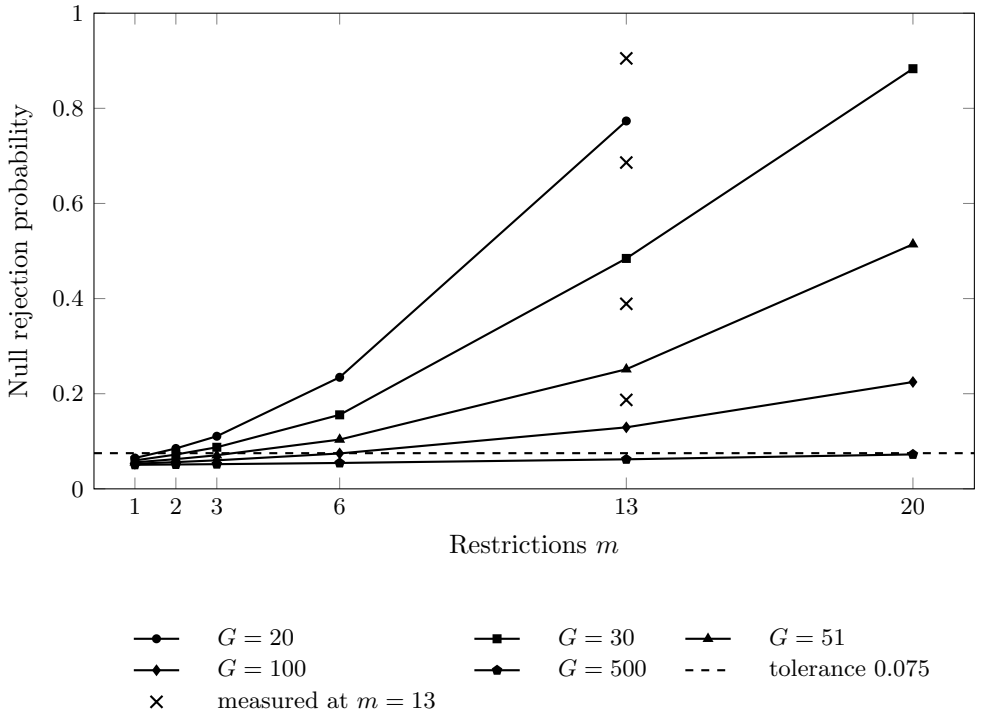


Figure 1: Exact size of the nominal 5% chi square test against the number of restrictions

Note: each curve gives the rejection probability of the nominal level-0.05 chi square test of the joint null $H_0 : \gamma_P = 0$ in (4.2), computed from (5.2) under Assumptions 2 and 3 at $\omega_2 = 1$, against the number of restrictions m, at five cluster counts G. The dashed line marks a tolerance of 0.075. The crosses are the measured rejection rates at $m = 13$ and $C = 5$ from Table 2, at $G = 20$, 30, 51 and 100 reading down, over 2,000 replications each with a Monte Carlo standard error of at most 0.011. They lie above the corresponding curves because the simulated design gives G_{eff} between $0.7142\,G$ and $0.7273\,G$ rather than the equal cluster weights of Assumption 3(ii). Sources: `results/inference/anatomy_expansion.csv`, `results/inference/size_table.csv` and `results/inference/anatomy_effective_clusters.csv`.

S7 The exact size against the number of restrictions

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